Machine Learning for Spectral Analysis

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1. Introduction

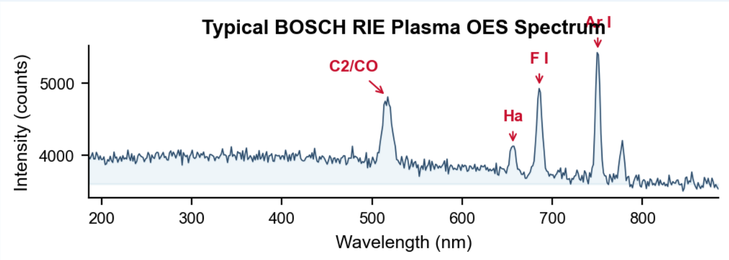
Optical Emission Spectroscopy (OES) provides **non-invasive, real-time diagnostics** of plasma processes by analysing emission spectra from excited species. Plasma-based manufacturing (semiconductor etching, surface treatment, thin-film deposition) relies on OES for process monitoring, but manual spectral interpretation is slow and error-prone.

This project develops an **automated ML pipeline** for plasma OES analysis targeting four objectives:

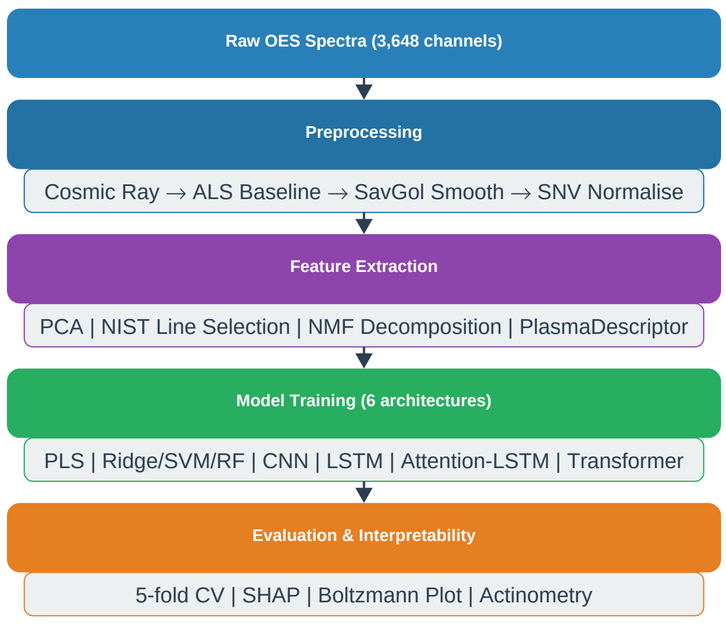
1. **Spectral feature identification** — automated peak detection, NMF decomposition, NIST database matching
2. **Plasma species classification** — SVM, RF, CNN, Transformer comparison (6 models)
3. **Spatiotemporal evolution** — Attention-LSTM phase prediction, species time-series extraction
4. **Semi-quantitative intensity** — actinometry (Coburn & Chen 1980), Boltzmann Te estimation

Three public datasets:

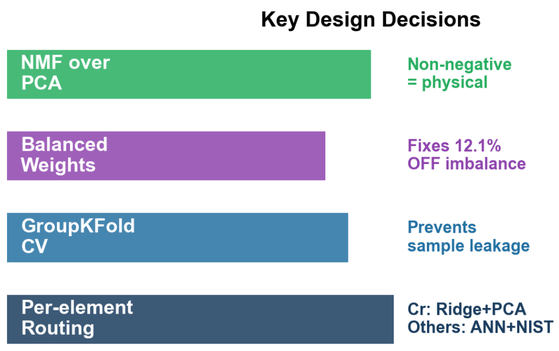
Dataset	Channels	Notes
LIBS Benchmark	40,002 ch	Steel composition (12 classes)
Mesbah CAP	51 ch	N2 plasma T_rot / T_vib
BOSCH RIE	3,648 ch	25 Hz, 10 days, SF6/C4F8



2. Methodology



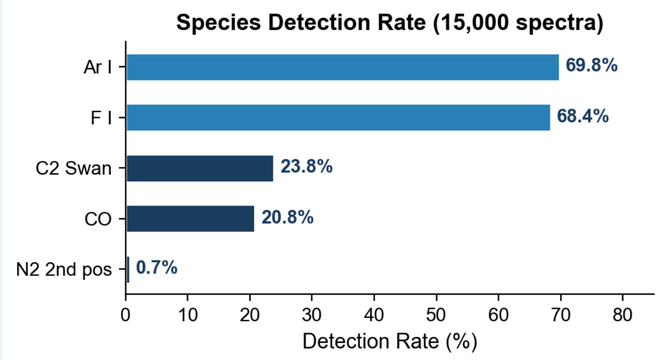
Hyperparameter optimisation: Optuna two-stage search (20 trials per target)



3. Species Identification

Non-negative Matrix Factorization (NMF) decomposes the spectral matrix **X** ≈ **W** · **H**, where each row of **H** is a pure-species emission spectrum and **W** contains the corresponding concentrations. NMF is physically appropriate because emission intensities are inherently non-negative.

Automated NIST matching: Each detected peak is compared against 39 reference emission lines from 13 species. The algorithm selects the closest database match within +/-1.5 nm tolerance. Species with peak intensity > μ + 3σ (global spectrum statistics) are classified as *present*.



Species	Lines (nm)	Origin
F I	685.6, 690.2, 703.7	SF6 etchant radical
Ar I	696.5, 750.4, 763.5	Carrier gas reference
C2 Swan	473.7, 516.5, 563.6	C4F8 carbon indicator
CO	451.1, 519.8, 561.0	C4F8 + O2 product
Si I	250.7, 252.4, 288.2	Si etch product
SiF	440.0, 442.5	Si + F recombination

Validation: NMF Component 0 peaks at 684.4 nm (≈ F I 685.6); Component 2 at 515.1 nm (approx. C2 Swan 516.5) — unsupervised decomposition confirms NIST species independently.

4. Classification Results

94.2% Accuracy (SVM/RF, 5-fold CV)

Model	Accuracy	F1 macro
SVM (RBF)	94.2%	0.843
Random Forest	94.2%	0.843
CNN (1D-Conv)	93.2%	0.822
Transformer	92.5%	0.802
Attention-LSTM	74.4%	—

Class	Precision	Recall	F1
Plasma OFF	0.87	0.61	0.717
Plasma ON	0.95	0.99	0.968

Species Detection Rates (13 species, 15,000 spectra):

Species	Detection	Species	Detection
Ar I	69.8%	C2 Swan	23.8%
F I	68.4%	CO	20.8%
N2 2pos	0.7%	Si I	0.4%

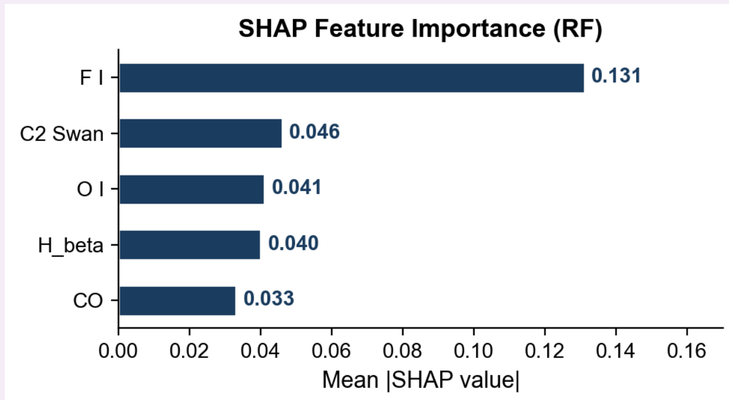
Model Architectures:

- **SVM/RF:** StandardScaler → Classifier pipeline. RF uses 200 trees; SVM uses RBF kernel (C=10, γ=scale). Both with balanced class weights.
- **CNN:** 3-layer Conv1D (32→64→128 channels, kernel 7/5/3) → AdaptiveAvgPool → FC(64) → Dropout(0.3) → output. Weighted CrossEntropyLoss, mini-batch training.
- **Transformer:** ViT-style 1D patch embedding (patch=64, d=128, 4 heads, 3 layers). [CLS] token classification. AdamW + cosine LR schedule.
- **Attention-LSTM:** 2-layer LSTM (hidden=64) → additive attention → FC. Trained on PCA(20) sliding windows (seq_len=10).

Analysis:

- **Traditional ML > Deep Learning:** SVM/RF achieve 94.2% vs CNN 93.2% and Transformer 92.5%. The plasma ON/OFF decision boundary is near-linear in feature space, giving kernel methods the advantage. Deep learning requires larger datasets to learn comparable representations.
- **Class imbalance impact:** Plasma OFF is only 12.1% of data. Despite balanced class weights, OFF recall = 0.61 (39% missed). The model is conservative in predicting OFF states, preferring high-confidence ON predictions (recall = 0.99).
- **Attention-LSTM underperforms (74.4%):** Temporal sequence classification on PCA(20) sliding windows loses spectral detail. Direct spectral classification (SVM/RF on 3,648 channels) retains full wavelength information, explaining the 20% accuracy gap.
- **Species detection validates physics:** Ar I (69.8%) and F I (68.4%) are dominant — consistent with SF6/Ar process gas. C2 Swan (23.8%) only appears during C4F8 passivation steps.

5. Interpretability & Physics



F I (fluorine radical) identified as the most discriminative species — consistent with its role as the primary etchant in SF6 plasma.

Root Cause Discovery: Etch/passivation spectral difference < 0.35σ (indistinguishable). RF power ON/OFF provides 0.56σ separation. Classification improved 74% → 94%.

Boltzmann T_exc: Excitation temperature = **13,334 K** (Boltzmann plot, 6 Ar I lines, 696.5–772.4 nm). Estimated via linear regression of ln(I·λ/gA) vs E_upper.

Why F I dominates: In SF6 plasma, electron-impact dissociation produces F radicals (SF6 + e- → SF5 + F + e-). The F I 703.7 nm line (2p4 3p → 2p4 3s transition, upper state 14.5 eV) has high transition probability (A = 6.4 x 10^7 /s) and is well-separated from neighbouring lines. Its intensity directly tracks F radical density, making it the most sensitive probe of etch chemistry.

Actinometry: Species concentration proportional to I_target / I_Ar (Coburn & Chen 1980). Ar carrier gas at known constant flow serves as reference. F/Ar ratio = 1.07 +/- 0.04; C2/Ar ratio = 1.13 +/- 0.20 (largest variability — reflects etch/passivation cycling).

Temporal analysis: Attention-LSTM achieves **74.4%** phase classification accuracy on PCA(20) embedding sequences. Attention weights reveal that transition timesteps between plasma states carry highest diagnostic information. DTW K-Means identifies 4 discharge phases (ignition, steady-state, transition, extinction) with 684 nm emission ratio > 2x between clusters.

6. Conclusions & Further Work

13 Species Detected
39 NIST lines, NMF validated

94.2% Accuracy | F1=0.843
6 models compared (SVM best)

Label Correction: 74% -> 94%
Data quality > model complexity

SHAP Validates Physics
F I = primary SF6 etchant

Future: Multi-class tracking | Real-time streaming | Transfer learning

Limitations:

- OES→process regression R2 < 0 (causal irreversibility)
- Boltzmann Te: narrow Ar I energy spread (0.33 eV)

References:

[1] Gidon *et al.* (2019) IEEE Trans. Radiat. Plasma Med. Sci.
[2] Coburn & Chen (1980) J. Appl. Phys.
[3] Lee & Seung (1999) Nature — NMF
[4] Lundberg & Lee (2017) NeurIPS — SHAP
[5] Vaswani *et al.* (2017) NeurIPS — Transformer
[6] Contreras *et al.* (2024) Anal. Chem.
[7] BOSCH dataset: Zenodo #17122442